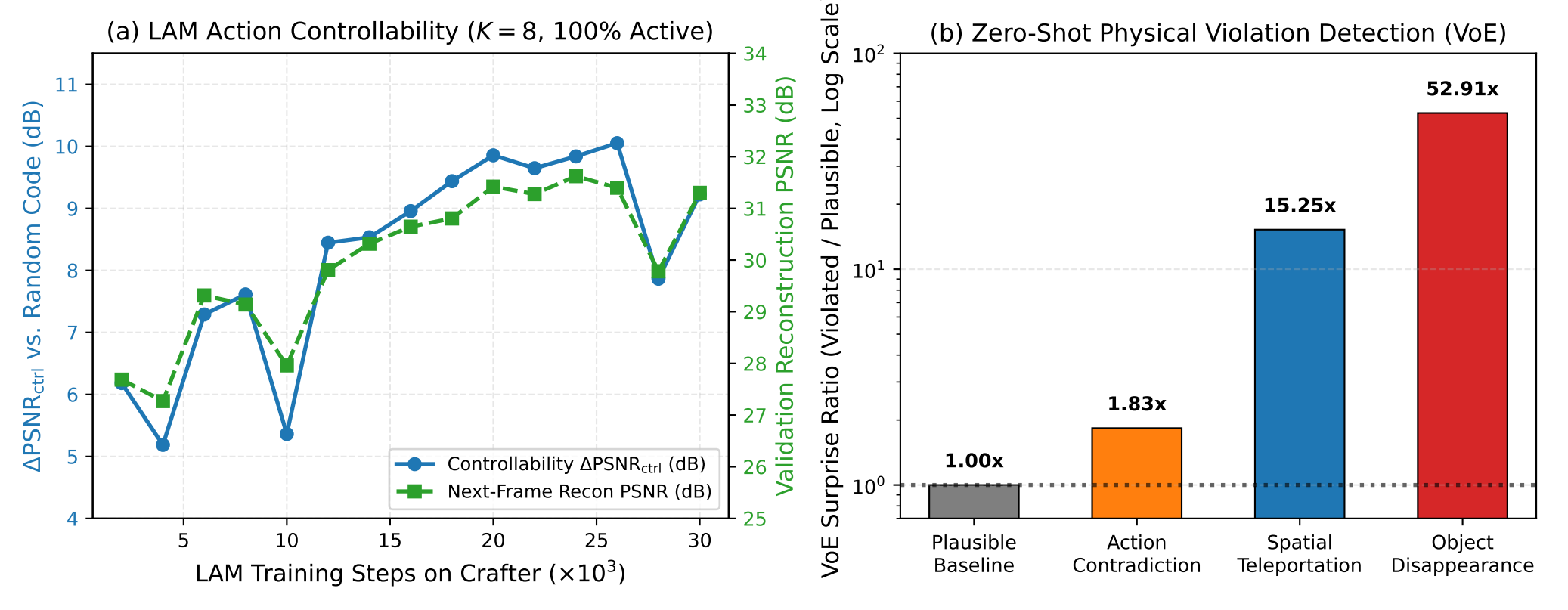
(a) LAM Action Controllability ($K = 8$, 100% Active)
$\Delta PSNR_{ctrl}$ vs. Random Code (dB)
Validation Reconstruction PSNR (dB)
LAM Training Steps on Crafter ($\times 10^3$)
Controllability $\Delta PSNR_{ctrl}$ (dB)
Next-Frame Recon PSNR (dB)
(b) Zero-Shot Physical Violation Detection (VoE)
VoE Surprise Ratio (Violated / Plausible, Log Scale)
1.00x
1.83x
15.25x
52.91x
Plausible Baseline
Action Contradiction
Spatial Teleportation
Object Disappearance